

Introduction to Professional Practices in IT

CS4001 – Professional Practices in IT

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FINANCING A START-UP COMPANY

Why Startups Need Capital?

The Basic Problem

- Customers pay **after delivery** → cash gap.
- Startups must spend on tools, staff, and living costs **before revenue arrives**.
- Without upfront money, survival is difficult.
Example: A freelancer needs a laptop, software licenses, internet subscription before even landing the first client.

IT Service Startup

Website-building company.

- Clients (other companies) pay late – 2 to 3 months minimum.
- Startups must cover salaries and rent until payment arrives.

Cash buffer for at least 3 months is essential.

Product Development Startup

If building a new software product.

- No revenue during development.

- Cash needed for:

Founders' and staff salaries.

Office costs (rent, utilities).

Equipment/software.

Marketing and promotion.

Travel, stationery, misc.

Loan interest (if borrowed). Even after launch, sales take months to stabilize.

Why a Business Plan?

- A roadmap for the company.
- A communication tool for investors/funders.
- Shows feasibility, market, competition, and financial forecasts.

Business Plan Document

- A description of **what the company will be doing**, together with information to show that it is technically feasible and that the founders of the company have the necessary expertise.
- A description of the market the company is aiming at, an **estimate of its size**, and an assessment of the competition.
- A prediction of the **financial performance** of the company. This will include budgets, cash flow predictions, and projected balance sheets and profit and loss accounts.

Sources of Finance

Every startup needs money to grow.

Governments encourage small businesses → provide different funding routes. All finance sources can be grouped into 3 categories:

1. Grants
2. Loans
3. Sale of Equity

Grants

- Free money, usually from government or charities.
- No repayment, but conditions apply (specific industries, job creation, location).
- Best for: Early R&D or social-impact projects.

Ignite – National Technology Fund through its network of **National Incubation Centers (NICs)** across cities like Islamabad, Lahore, Karachi, and Peshawar, Ignite provides early-stage tech startups with **seed grants, office space, mentorship**, and investor connections

Loans

- Borrowed money → must be repaid + interest.
- Types:
 - Overdraft: Flexible, pay interest only on what you use.
 - Long-term loan: Fixed repayments, predictable.
- Risks: Needs security .Bank may withdraw overdraft unexpectedly.
- Best for Businesses with steady cash flow.

Overdraft

- ABC Traders had to **pay salaries** and **restock inventory**, but their customers haven't paid yet, and their bank balance is only \$2,000.
They have an **overdraft facility** of **\$10,000**.
- They withdraw an extra **\$5,000** using the overdraft.
- Interest is charged **only** on the \$5,000 they used, not the full \$10,000.
- They repay it within 2 weeks when customer payments come in.
- This helps **manage short-term cash flow** without taking a full loan.

Long Term Loan

- ABC Traders wants to **open a new store** and needs **\$50,000** for renovation, furniture, and equipment.
- They apply for a **5-year term loan** from the bank.
- The bank approves the loan with a **fixed interest rate**.
- ABC Traders repays the loan in **monthly installments** over 5 years.
- This loan helps them **expand the business** and invest in long-term assets.



Feature	Overdraft	Long-Term Loan
Amount	\$5,000 (from a \$10,000 limit)	\$50,000
Purpose	Salary/inventory (short-term)	Business expansion (long-term)
Duration	2 weeks	5 years
Repayment	Repaid when funds available	Fixed monthly payments
Interest	On \$5,000 for 2 weeks	On full \$50,000 for 5 years



Sale of Equity

- Equity = ownership share in the company.
- When a startup or company needs money, instead of borrowing (like loans), it can sell part of its ownership to investors in exchange for cash.
- These investors become shareholders. Unlike a loan, the company doesn't have to repay the money, but it has to share profits, decision-making power, and sometimes control.

How it Works

- Company valuation is determined (how much the company is worth).
- A percentage of ownership (shares) is offered to investors.
- Investors give money in exchange for those shares.
- The company uses this money to grow (hire staff, develop products, expand to new markets).
- If the company succeeds, both founders and investors benefit because the value of shares increases.

Types of Equity Investors

Business Angels – wealthy individuals investing their own money. A retired IT professional in Karachi investing PKR 5 million into a promising fintech startup.

Venture Capitalists (VCs) – investment firms that invest in startups with high growth potential. Pakistani VC firm Lakson Venture Capital (LVC) investing in Techlogix or health-tech startups.

Gearing

The relationship between loan capital (borrowed money) and equity capital (owners' money). Shows how dependent a company is on borrowed funds.

Risks of High Gearing

If company fails → lenders get paid first, shareholders may lose everything.

- High interest burden reduces flexibility.
- Lenders reluctant to give more money if gearing is too high.

Companies must balance:

- Using **loans** (to grow faster).
- Using **equity** (safer, but ownership gets diluted).

Balance Sheet

The purpose of the balance sheet is to show what the company owns – its **assets** – and what it owes – its **liabilities**. It is a snapshot of the state of the company at a particular point in time, normally at the end of the last day of the company's financial year.

Scenario 1

Jemima is a university student who decided to check her financial position at the end of 2021, much like how a company prepares a balance sheet. When she looked at her assets, she found she only had £25 in cash (slightly less than last year's £40) but her bank balance improved to £361 compared to £240 the previous year. She had prepaid £1,300 for her accommodation, a little higher than last year's £1,180, and a friend owed her £18. Her computer, once worth £360, had gone down in value to £240, and her guitar had also dropped slightly in value from £180 to £160. Personal items stayed the same at £200.

Scenario 1

On the other hand, Jemima's liabilities had grown much faster. Her credit card bill increased from £64 last year to £174 this year, and most importantly, her student loan had jumped significantly from £1,900 to £4,800. Jemima's wants to know how much total assets she has in 2021 as compared to last year. Also, she needs to find her net worth as of 2021 with reference to 2020.

Table 6.1 Balance sheet for a student**Jemima Puddleduck****Balance sheet**

As at 31 October 2021	2021	2020
ASSETS		
Cash in hand	25	40
Cash at bank	361	240
Pre-paid accommodation	1,300	1,180
Debts owed by friends	18	0
Computer	240	360
Guitar	160	180
Personal effects	200	200
<i>Total assets</i>	2,304	2,200
LIABILITIES		
Credit card bill	174	64
Student loans	4,800	1,900
<i>Total liabilities</i>	4,974	1,964
<i>Net worth</i>	(2,670)	236

Scenario 2

XYZ Software Ltd, a mid-sized services company, prepared its balance sheet for October 2021. The company noticed several interesting changes compared to the previous year. On the asset side, the company had invested in new intangible assets worth £475 possibly software licenses or intellectual property whereas the previous year showed none. Tangible assets like office equipment and infrastructure rose slightly to £960 from £770. However, investments dropped to £50 compared to £82 the year before, showing they might have cashed out some earlier investments.

Scenario 2

Current assets were also strong. Work in progress in this year is £550 as compared to last year that was £621, and debtors' money owed by clients jumped to £3,400 from £2,580, which indicates more projects delivered but not yet paid for. Cash at hand and in the bank improved significantly to £2,491 compared to £1,770 last year. But the company also faced rising liabilities. Short-term creditors (amounts due within one year) increased to £3,210 from £2,601. Borrowings also rose slightly to £154 from £61, though still a manageable figure. Provisions for liabilities and charges are £7 in 2021, whereas last year it was £16. After accounting for these obligations what will be the company's net assets in comparison with two years?

Scenario 2

Looking at capital and reserves, XYZ Software had increased its share capital slightly (to £318 from £308) and raised its share premium reserve to £350 from £145, meaning they likely issued shares at a premium. What will be profit and loss status of this company in comparison of two years?

FIXED ASSETS		
Intangible assets	475	–
Tangible assets	960	770
Investments	50	82
<i>Total fixed assets</i>	1,485	852
CURRENT ASSETS		
Work in progress	550	621
Debtors	3,400	2,580
Cash in hand and at bank	2,491	1,770
<i>Total current assets</i>	6,441	4,971
CREDITORS: AMOUNTS FALLING DUE WITHIN ONE YEAR	(3,210)	(2,601)
<i>Net current assets</i>	3,231	2,370
Total assets less current liabilities	4,716	3,222
CREDITORS: AMOUNTS FALLING DUE AFTER ONE YEAR		
Borrowings	(154)	(61)
Provisions for liabilities and charges	(7)	(16)
<i>Net assets</i>	4,555	3,145
CAPITAL AND RESERVES		
Called-up share capital	318	308
Share premium reserve	350	145
Profit and loss account	3,887	2,692

Scenario 3

Jemima Puddleduck, a university student, reviewed her finances for the year ending October 2021. Her total income came from two main sources: her parents contributed £1,500, and she earned £1,840 from a summer job. Compared to the previous year's £2,982, her income slightly improved. However, her expenses were far higher than her income. The largest cost was her hall fees, which amounted to £2,100, up from £1,980 last year. Her course fees came to £1,050, that was £25 higher than year with additional smaller but necessary expenses like £30 on books that were £5 higher, £179 on clothes that were £59 higher as compared to previous year and personal items, and last year transport expense was £112 that was increased this year with £22.

Scenario 3

Basic living costs also added up significantly. She spent £1,400 on food and £1,303 on entertainment, both of which rose compared to the previous year. Previous year it was £1,247 on food and £840 on entertainment. Additionally, depreciation on her belongings, such as her laptop, was estimated at £140, the same as the previous year. Calculate the excess of income over expenditure.

Table 6.3 Income and expenditure account for a student

Jemima Puddleduck

Income and expenditure account

Year ended 31 October 2021	2021	2020
INCOME		
Contribution from parents	1,500	1,300
Income from summer job (net)	1,840	1,682
<i>Total income</i>	3,340	2,982
EXPENDITURE		
Course fees	1,050	1,025
Hall fees	2,100	1,980
Books	30	25
Clothes and personal items	179	120
Transport	134	112
Food	1,400	1,247
Entertainment	1,303	840
Depreciation	140	140
<i>Total expenditure</i>	6,336	5,489
<i>Excess of income over expenditure</i>	(2,996)	(2,507)

Scenario 4

XYZ Software Ltd, a medium-sized IT services company, has just closed its financial year ending 31 October 2021. Compared to the previous year, the company experienced strong growth. Revenue from its continuing operations rose from £11,001 in 2020 to £14,311 in 2021, and an additional £407 came from a recent acquisition.

But with growth came challenges. The cost of sales increased sharply, moving from £8,699 to £11,604 reflecting higher staff salaries, outsourcing expenses, and infrastructure costs needed to deliver bigger projects.

Scenario 4

Administrative and operating costs also grew, from £805 in 2020 to £1,177 in 2021, due to expansion and additional marketing for new clients. The company managed its debts well, reducing interest payments slightly from £27 to £23. After tax, however, the burden increased: £719 paid in 2021 compared to £480 in 2020.

What will be the total retained profit in 2021 as compared to 2020.

Table 6.4 Profit and loss account for a services company

XYZ Software Ltd

Profit and loss account	2021	2020
Year ending 31 October 2021	£'000	£'000
TURNOVER		
Continuing operations	14,311	11,001
Acquisitions	407	–
<i>Total turnover</i>	14,718	11,001
COST OF SALES	(11,604)	(8,699)
<i>Gross profit</i>	3,114	2,302
OTHER OPERATING EXPENSES	(1,177)	(805)
<i>Operating profit</i>	1,937	1,497
INTEREST PAYABLE	(23)	(27)
<i>Profit on ordinary activities before taxation</i>	1,914	1,470
TAX ON PROFIT ON ORDINARY ACTIVITIES	(719)	(480)
<i>Retained profit for the year</i>	1,195	990

Scenario 5

Jemima Puddleduck, a university student, reviewed her cash flow statement for the year ending 31 October 2021. Over the year, she received an additional £2,900 from her student loan, which was higher than the £1,900 she had taken the previous year. On top of that, she accounted for £140 of depreciation on her belongings, the same as the previous year. On the outflow side, Jemima's spending was significant. Her day-to-day living, education, and other expenses amounted to £2,996, slightly higher than last year's £2,507. Additionally, she lent £18 to a friend, which further reduced her available cash. What will be the total Cash Inflow, Cash Outflow and net increase/decrease in both years?

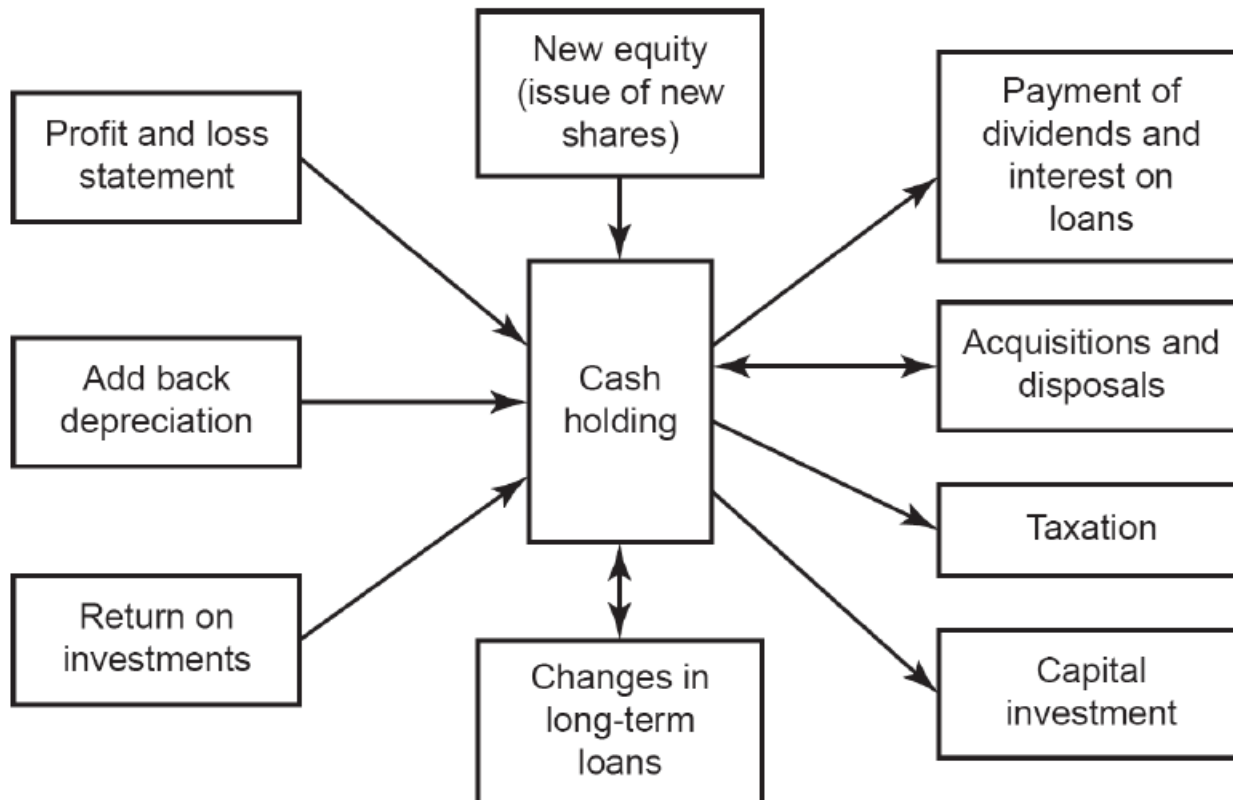
Table 6.5 Cash flow statement for a student

Jemima Puddleduck

Cash flow statement

Year ended 31 October 2021	2021	2020
CASH INFLOW		
Addition to student loan	2,900	1,900
Add back depreciation	140	140
<i>Total cash inflow</i>	3,040	2,040
CASH OUTFLOW		
From income and expenditure account	2,996	2,507
Loans made to friends	18	0
<i>Total cash outflow</i>	3,014	2,507
<i>Increase/(decrease) in cash over the year</i>	26	(467)

Sources and destinations of cash flows



Scenario 6

XYZ Software Ltd has shared its cash flow statement for the year ending 31 October 2021. The company generated a net cash inflow of £2,105 from its operating activities, up from £1,620 in 2020, showing stronger business performance. However, several significant outflows reduced this figure. The company spent £220 on capital expenditure and financial investments which were £265 last year, paid £719 in taxes in 2021 which were £480 in 2020, and recorded £380 in acquisitions and disposals in the year 2021. In addition, £23 went toward servicing finance, and dividends were distributed to shareholders which was £27 last year.

Scenario 6

To strengthen its financial position, XYZ Software Ltd raised £215 by issuing new share capital, which was more than double the £100 raised in 2020. On the other hand, it repaid £50 of long-term loans during the year. Calculate total cash outflow before financing and net cash inflow before financing. Also, calculate net cash inflow after financing and increase in cash both years.

Table 6.6 Cash flow statement for a software company

XYZ Software Ltd

Cash flow statement	2021	2020
Year ending 31 October 2021	£'000	£'000
Net cash inflow from operating activities	2,105	1,620
Returns on investments and servicing of finance	(23)	(27)
Capital expenditure and financial investment	(220)	(265)
Taxation	(719)	(480)
Acquisitions and disposals	(380)	
Equity dividends paid		
<i>Cash outflow before financing</i>	<i>(1,342)</i>	<i>(772)</i>
<i>Net cash inflow before financing</i>	763	848
FINANCING		
Issue of share capital	215	100
Repayment of long-term loan	(50)	
<i>Net cash inflow from financing</i>	<i>165</i>	<i>100</i>
<i>Increase in cash in the year</i>	928	948

Class Activity 5

<https://tinyurl.com/Fall2025-Activity5>

Question 1

Profit and Loss Account (Income Statement)

For the Year Ended [Date]

Description	Amount
Revenue	\$500,000
Expenses:	
Salaries	\$150,000
Marketing	\$100,000
Other Expenses	\$100,000
Total Expenses	
Net Income	

Question 2

Balance Sheet

Assets	Amount
Current Assets:	
Cash	\$50,000
Accounts Receivable	\$50,000
Total Current Assets	
Non-Current Assets:	
Equipment	\$150,000
Total Assets	

Liabilities and Equity	Amount
Current Liabilities:	
Accounts Payable	\$100,000
Other Liabilities	\$20,000
Total Current Liabilities	
Long-Term Liabilities:	
Long-Term Loan	\$50,000
Total Liabilities	
Equity:	
Initial Funding	\$200,000
Retained Earnings (Net Income)	\$(150,000)
Total Equity	
Total Liabilities and Equity	

Profit and Loss Account (Income Statement)

For the Year Ended [Date]

Description	Amount
Revenue	\$500,000
Expenses:	
Salaries	\$150,000
Marketing	\$100,000
Other Expenses	\$100,000
Total Expenses	\$(350,000)
Net Income	\$150,000

Balance Sheet

Assets	Amount
Current Assets:	
Cash	\$50,000
Accounts Receivable	\$50,000
Total Current Assets	\$100,000
Non-Current Assets:	
Equipment	\$150,000
Total Assets	\$250,000

Liabilities and Equity	Amount
Current Liabilities:	
Accounts Payable	\$100,000
Other Liabilities	\$20,000
Total Current Liabilities	\$120,000
Long-Term Liabilities:	
Long-Term Loan	\$50,000
Total Liabilities	\$170,000
Equity:	
Initial Funding	\$200,000
Retained Earnings (Net Income)	\$(150,000)
Total Equity	\$50,000
Total Liabilities and Equity	\$220,000